Lab 1

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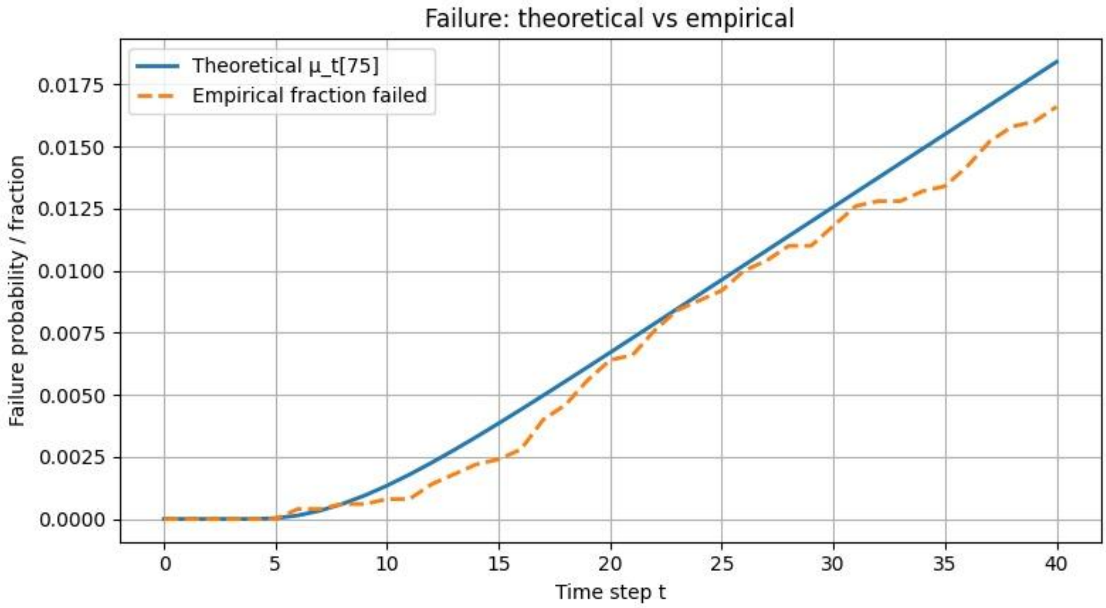
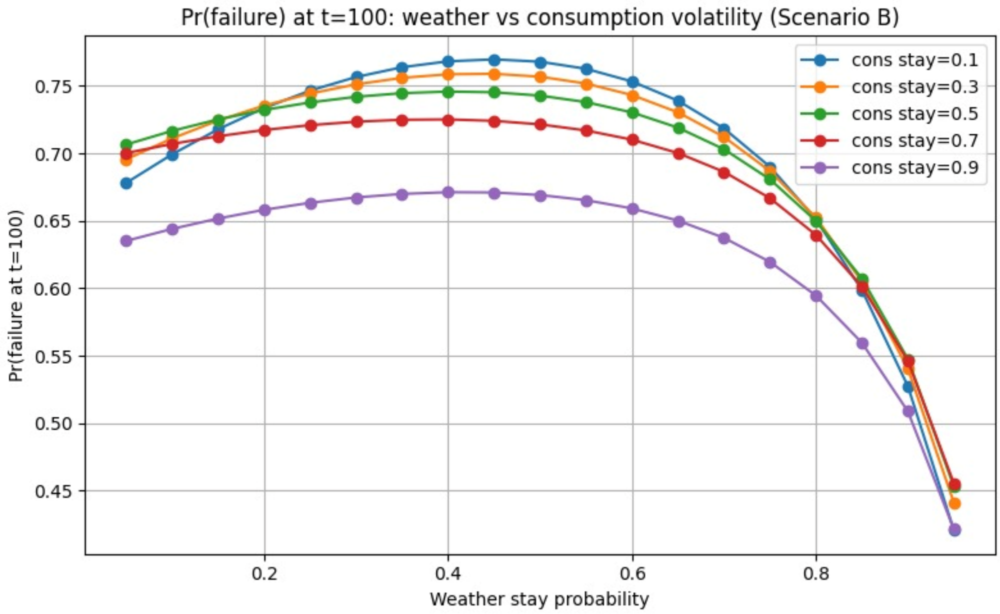
1- A basic method would be to sum all probabilities of each line to see if they all sum to one, if even a single line does not follow this rule the matrix is wrong.
2- Generating a theroretical value is just starting with the inicial position vector and then in each state multiply that by the transition matrix, the empirical is using randomness to get the values in question, which gave us this:



3- Given enough time the Iterative method converges to the stationary distribution, this because they are theoretically the same, meaning the stationary distribution is the probability distribution on time step t infinite, and the Iterative method if infinite is the probability distribution.
4- The stationary distribution on scenario C is trivial as we have a state from where we cannot go anywhere, but we can go to that state through others, then given enough time the probability of being on that state converges to 1, making a better quantitative measure the time before that happens, which in our program accumulates to thousands of time steps, around 7000.

5- In changing the stay values of both consumption and weather we can conclude that the failure probability converges to 1 the fastest the more distributed weather is, around 0.5 and the lower C is.



6- The homeowner should aim for a battery level 2 as this does not have direct transition to failure but 4 has, correlating to a better probability and longer time-span:

B=2 TTF: min=4.80 max=9.85 mean=6.61

B=4 TTF: min=3.64 max=7.25 mean=4.64

TTF: Time To Failure